Martin A. Miguel

Post-doctoral Fellow

Canada

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Academic Positions

Western University and McGill University

Post-doctoral Fellowship — *September 2025 - Present*

- Advisor: Jessica Grahn - Music and Neuroscience Lab, Psychology Department, Western University - London, Ontario, Canada
- Advisor: Robert Zatorre - Auditory Cognitive Neuroscience Lab - Montreal Neurological Institute, McGill University - Montreal, Quebec, Canada

McMaster University — *Hamilton, Ontario, Canada*

Post-doctoral Fellowship — *September 2022 - August 2025*

- Advisor: Laurel Trainor - Auditory Development Lab & LIVELab, McMaster University
- Advisor: Jonathan Cannon - METRE Lab, McMaster University

Education

University of Buenos Aires (UBA) — *Buenos Aires, Argentina*

PhD in Computer Science — *April 2016 - August 2022*

- Advisor: Diego Fernandez Slezak - Applied Artificial Intelligence Lab (LIAA), Computer Science Department, University of Buenos Aires, Buenos Aires, Argentina; Computer Science Institute, National Scientific and Technical Research Council (CONICET)-UBA, Argentina
- Co-advisor: Mariano Sigman - Neuroscience Laboratory, Torcuato Di Tella University, Buenos Aires, Argentina; Faculty of Language and Education, Nebrija University, Madrid, Spain

Contemporary Music School — *Buenos Aires, Argentina*

Professonal Musician — *April 2015 - June 2017 (Paused)*

University of Buenos Aires — *Buenos Aires, Argentina*

BS + MS in Computer Science — *April 2008 - December 2015*

Main Publications

Cossavella, F., Dottori, M., **Miguel, M. A.**, Fernandez Slezak, D. (2025) Alpha band activity over the sensorimotor cortex during passive music listening correlates with beat tapping performance. Proceedings of the Annual Meeting of the Cognitive Science Society, 47. Conference paper.

Somacal, L., Riera, P., Fernandez Slezak, D. **Miguel, M. A.** (2024) Symbolic music style transfer via latent space transformations: model and evaluation. Latin American Music Information Retrieval Workshop, Rio de Janeiro, Brazil, December 2024. Conference paper on machine learning model for music style transfer in symbolic music.

Cossavella, F., **Miguel, M.A**, Fernandez Slezak, D. The Role Of The Motor System In The Processing Of Rhythmic Complexity: A Critical Review. (2024) CogSci 2024, Rotterdam, Netherlands, July 2024. (e-ISSN: 1069-7977) `https://escholarship.org/uc/item/5vx1w3qg` Conference paper reviewing fMRI studies reporting the relative activation of motor areas with respect to rhythmic complexity.

Kiss L, Guiot C, Hashim S, D'Aleman Arango N, **Miguel MA**. (2022) The 14th International Conference of Students of Systematic Musicology (SysMus21). Music & Science. doi:10.1177/20592043221076613. Conference report.

Miguel, MA and Fernandez Slezak, D. (2021) Modeling beat uncertainty as a 2D distribution of period and phase: a MIR task proposal. Proc. of the 22nd Int. Society for Music Information Retrieval Conf., Online. (ISBN 978-173272990-2) Paper describing a methodology to model beat uncertainty considering period and phase from free tapping data and an evaluation criterion for MIR models.

Pironio N, Fernandez Slezak D, and **Miguel MA**. (2021) Pulse clarity metrics developed from a deep learning beat tracking model. Proc. of the 22nd Int. Society for Music Information Retrieval Conf., Online, 2021. (ISBN 978-173272990-2) Paper describing metrics of pulse clarity obtained from modifications to a neural-network based beat tracking model.

Miguel MA, Riera P, and Fernandez Slezak D. (2021) A simple and cheap setup for timing tapping responses synchronized to auditory stimuli. Behav Res. https://doi.org/10.3758/s13428-021-01653-y Paper describing an experimental setup for capturing timing of tapping responses synchronized against auditory stimuli. The setup requires minimal programming skills and uses unexpensive equipment.

Miguel MA, Sigman M, and Fernandez Slezak D. (2020) From beat tracking to beat expectation: Cognitive-based beat tracking for capturing pulse clarity through time. PLoS ONE 15(11): e0242207. https://doi.org/10.1371/journal.pone.0242207 Paper presenting a model of beat tracking adapted to produce a metric of pulse-clarity over time.

Other publications

Vladisauskas, M., Paz, G.O., Nin, V., Guillén, J.A., Belloli, L., Delgado, H., **Miguel, M.A.;**, Macario Cabral, D., Shalom, D.E., Forés, A., et al. (2024) The Long and Winding Road to Real-Life Experiments: Remote Assessment of Executive Functions with Computerized Games—Results from 8 Years of Naturalistic Interventions. Brain Sci. 2024, 14, 262. `https://doi.org/10.3390/brainsci14030262` Review of 8 years of usage of the MateMarote platform to test the efficacy of computerized games used in naturalistic settings to train attention and excecutive function in children.

Belloli, L. **Miguel, M.A.**, Goldin, A.P. (2016) Mate Marote: a BigData platform for massive scale educational interventions. 45-JAIIO, 2016, Buenos Aires, Argentina (ISSN: 2451-7569, p107-114). Paper describing a web platform that hosts and collects data from educational games.

Teaching Experience

Teaching Fellow, Algorithms and Data Structures II, *University of Buenos Aires* — April 2016 – August 2022
Teaching Assistant, Algorithms and Data Structures II, *University of Buenos Aires* — March 2011 – July 2012

Industry Experience

Technical Consultant, *MateMarote Project (Online Educational Games)* — June 2017 – June 2022
Data Scientist, *Avenida.com* — January 2016 – March 2016
Software Engineer, *MateMarote Project (Online Educational Games)* — April 2015 – December 2015
Software Engineer Intern, *Google.com* — January 2014 – April 2014
Java Programmer, *Despegar.com* — August 2012 – December 2013
Java Programmer (J2ME / Blackberry), *SenseByte* — January 2009 – January 2010

Mentoring

Laurine Andraws, Mentor on student research project: student assisted with data collection in experiment of sensorimotor synchronization with distractors — Winter 2025
Psychology, Neuroscience and Behavior department, McMaster University, Hamilton, Ontario, Canada

Sophie Cournoyer, Vanessa Vandewetering, Laura Corrigal, Katerine Bilous, Mentor on student research project: students assisted with data collection in experiment of partner dancing using motion capture — Winter 2025
Psychology, Neuroscience and Behavior department, McMaster University, Hamilton, Ontario, Canada

Blinar Fuentes Trejos, Koko Kumazawa, Isabel Antonelli, Mentor on student research project: students assisted with data collection in experiment of partner dancing using motion capture — Winter and Fall 2024
Psychology, Neuroscience and Behavior department, McMaster University, Hamilton, Ontario, Canada

Sahejpreet Dhillon, Mentor on student research project: student assisted with data collection in experiment of sensorimotor synchronization with distractors — Winter 2024
Psychology, Neuroscience and Behavior department, McMaster University, Hamilton, Ontario, Canada

Francisco Cosvella, Assistant advisor in PhD: role of motor areas in rhythmic perception, neurological evidence and computational modelling — 2023 - Ongoing
PhD in Psychology, Psychology Department, Faculty of Social Sciences, University of Buenos Aires, Argentina.

Lucas Somacal, Main advisor in master's thesis: Modeling musical style with Variational Autoencoders — 2023
Licenciature in Computer Science, Computer Science Department, Faculty of Natural and Exact Sciences, University of Buenos Aires, Argentina.

Francisco Cossavella, Co-advisor in master's thesis: Cortical mu-rhythms during beat perception and synchronization with naturalistic music stimuli. — 2021
Licenciature in Psychology, Faculty of Behavior and Human Sciences, Favaloro University, Buenos Aires, Argentina

Lucas Somacal, Mentor of undergraduate research internship: Exploration of music style transfer techniques based on VAE's latent spaces from symbolic music data. — 2021
Computer Science Department, Faculty of Natural and Exact Sciences, University of Buenos Aires, Argentina.

Nicolás Pironio, Mentor of undergraduate research internship: Analysis of the behaviour of a deep-learning beat tracking model to estimate pulse clarity. — 2020
Computer Science Department, Faculty of Natural and Exact Sciences, University of Buenos Aires, Argentina.

Conferences and Schools

Conference presentations

Miguel, M.A, Trainor, L., Cannon, J. *Inclination to synchronize with a sound is independent of its association with movement... or is it? New results from synchronizing with voice.* 20th Rhythm Perception and Production Workshop, Jyväskylä, Finland, 2025

Miguel, M.A, Cannon, J., Trainor, L. *Does switching partners enhances non-verbal communication in partner dancing?* 20th Annual Neuromusic Conference, Hamilton, Canada, November 2024

Kirk, R., Anderson, C., **Miguel, M.A**, Wood, E., Tawfik, H., Bosnyak, D., Trainor, L. *Highly expressive moments correspond to less audience synchronisation during a live concert.* 20th Annual Neuromusic Conference, Hamilton, Canada, November 2024

Kirk, R., Anderson, C., **Miguel, M.A**, Wood, E., Tawfik, H., Bosnyak, D., Trainor, L. *Expressivity, emotion, and synchronisation of subjective and neurophysiological responses during a live concert.* Workshop on Concert Research, Oslo, Finland & Online, September 2024

Miguel, M.A, Premraj, D., Hanna, A., Jalbert, J., Cannon, J. *Entrainment to audiomotor feedback: preliminary evidence and model* Society for Music Perception and Cognition, Banff, Canada, July 2024

Miguel, M.A, Cannon, J., Trainor, L. *Exploitation vs. exploration in partner dancing: whether switching partners enhances communication in partner dancing.* The Neurosciences and Music – VIII, Helsinki, Finland, June 2024

Miguel, M.A, Cannon, J., Trainor, L. *Exploitation vs. exploration in partner dancing: whether switching partners enhances communication in partner dancing.* 3rd Joint Seminar on Music, Sport and Health, Jyväskylä, Finland, June 2024

Miguel, M.A, Cannon, J., Trainor, L. *Exploitation vs. exploration in partner dancing: whether switching partners enhances communication in partner dancing.* Brains and Bodies in Social Interaction, Learning and Wellbeing, Jyväskylä, Finland, June 2024

Miguel, M.A, Trainor, L., Cannon, J. *Co-representation vs. attenuation: whether motor representation of a distractor makes it more distracting.* Brains and Bodies in Social Interaction, Learning and Wellbeing, Jyväskylä, Finland, June 2024

Miguel, M.A, Trainor, L., Cannon, J. *Co-representation vs. attenuation: whether motor representation of a distractor makes it more distracting.* Symposium on Timing and Rhythm (STAR), Hamilton, Canada, April 2024

Cossavella, F., **Miguel, M.A**, Fernandez Slezak, D. *The Role Of The Motor System In The Processing Of Rhythmic Complexity: A Critical Review* Neuromusic 19, Hamilton, Canada, October 2023

Miguel, M.A, Trainor, L., Cannon, J. *Co-representation vs. attenuation: whether motor representation of a distractor makes it more or less distracting.* Neuromusic 19, Hamilton, Canada, October 2023 (DOI 10.17605/OSF.IO/NC7FE)

Kirk et al. *The highs and lows of music: subjective and neurophysiological responses during a live concert experience.* International Conference of Students of Systematic Musicology 23, Online and Sheffield, United Kingdom, October 2023.

Miguel, M.A, Cannon J., Trainor, L. *Modeling the subjective beat in period, phase and uncertainty.* Neuromusic 18, Hamilton, Canada, November 2022 (DOI 10.17605/OSF.IO/2J6HM)

Miguel, M.A, Fernandez Slezak, D. *Modeling beat ambiguity in period and phase.* International Conference of Students of Systematic Musicology 21, Online and Aahrus, Denmark, November 2021 (DOI 10.17605/OSF.IO/5WRS3) Poster presenting a methodology from gathering a beat distribution from free tapping data.

Miguel, M.A, Sigman, M., Fernandez Slezak, D. *A continuous model of pulse clarity: towards inspecting affect through expectations in time.* Neuromusic VII, Online and Aahrus, Denmark, June 2021. Poster describing an updated evaluation of our beat expectation model's measure of pulse clarity considering new data and constrating models.

Pironio, N., Fernandez Slezak, D., **Miguel M.A.**. *Evaluation of pulse clarity models on multiple datasets.* Rhythm Perception and Production Workshop 2021, Online and Oslo, Norway, June 2021 (DOI 10.17605/OSF.IO/SDQ5P) Poster preseting the evaluation of pulse clarity models on multiple datasets.

Pironio, N., Fernandez Slezak, D., **Miguel M.A.**. *Analysis of the behaviour of a beat tracking model to estimate pulse clarity.* 16th International Conference on Music Perception and Cognition, Online, July 2021 Poster presenting metrics of pulse clarity obtained from modifications to a neural-network based beat tracking model.

Miguel, M.A, Sigman, M., Fernandez Slezak, D. *Experimental setup for exploring subjective tacti distribution and pulse clarity.* Biannual meeting of the Society of Music Perception and Cognition 2019, New York, USA, August 2019 (DOI 10.17605/OSF.IO/7SQAW). Poster describing a novel experimental setup that extends on previous methods allowing exploration of subjective tacti on top of pulse clarity.

Miguel, M.A, Sigman, M., Fernandez Slezak, D. *A continuous model of pulse clarity: towards inspecting affect through expectations in time.* Biannual meeting of the Society of Music Perception and Cognition 2019, New York, USA, August 2019 (DOI 10.17605/OSF.IO/FGVB2). Poster describing how our beat expectation model's measure of pulse clarity relates with pulse clarity extracted from empirical data.

Schools

Assistance to KHIPU 2019. University of the Republic, Montevideo, Uruguay

Assistance to Machine Learning Summer School (MLSS 2018). Torcuato Di Tella University, Buenos Aires, Argentina

Assistance and volunteering at International Joint Conference in Artificial Intelligence 2015. Buenos Aires, Argentina

Awards, Fellowships, & Grants

2016 - 2022	**PhD Scholarship**, National Scientific and Technical Research Council, Argentina	*$ 100,000*
2013	**"TIC" Scholarship for end of program**, Computer Science Department, University of Buenos Aires, Argentina	*$ 3,000*

Outreach & Professional Development

Science communication

2017, 2018, 2019	**University Program Promotion: Introduction to programming with music**, Designer and lecturer	*University of Buenos Aires, Buenos Aires, Argentina*
2018, 2019	**University Program Promotion: program talks**, Designer and lecturer	*University of Buenos Aires, Buenos Aires, Argentina*

Non-research activities

I collaborated in the redesign of the Computer Science Program at the University of Buenos Aires for the new program approved for 2023

cvpubs